

SolarNode v2.0: A 5G/6G-Native, AI-Powered Disaster Communication Mesh

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1. Project Title

SolarNode v2.0: A 5G/6G-Native, AI-Powered Disaster Communication Mesh

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2. Executive Summary

Natural disasters routinely destroy terrestrial communication infrastructure, leaving rescue teams blind and disorganized during the critical first 48 hours. In the 2023 Turkey-Syria earthquake, over 80% of mobile coverage was lost within 6 hours. Current solutions—satellite phones, walkie-talkies, and drones—are expensive, short-range, or power-limited.

SolarNode addresses this gap with a swarm of solar-powered, self-healing mesh nodes that deploy instantly via drone or helicopter. Each node integrates an ESP32 microcontroller, LoRa radio, GPS, and solar panel with battery storage. When dropped into a disaster zone, nodes autonomously form a resilient mesh network, relaying survivor GPS coordinates and priority messages to a cloud command center via 5G/6G Non-Terrestrial Network (NTN) gateway nodes.

Our enhanced v2.0 architecture integrates 5G/6G NTN for satellite backhaul, Device-to-Device (D2D) communication, and 6G Integrated Sensing and Communication (ISAC) for survivor detection. An AI-powered analytics engine (Isolation Forest) detects network anomalies in real-time, predicts node failures, and provides optimization recommendations.

Python simulations using `networkx` demonstrate 94.2% Packet Delivery Ratio (PDR), 142-hour network lifetime (2.5× longer than battery-only systems), and 95% coverage with redundant nodes. SolarNode reduces rescue "blind time" from 4–6 hours to under 30 minutes, costing under \$100 per unit and deployable globally for earthquakes, floods, and hurricanes.

3. General Description

The Communication Blackout Problem

When natural disasters strike, the first hours are characterized by a breakdown of communication. Cellular towers collapse, fiber-optic backhauls are severed, and power grids fail. Rescue teams are forced to rely on paper maps, word-of-mouth, and sporadic satellite phone calls—leading to delayed triage, duplicate efforts, and lost lives.

The "golden window" for disaster response is the first 48 hours. Yet without reliable communication, first responders cannot:

- Coordinate search-and-rescue operations across fragmented teams.
- Receive real-time GPS coordinates of survivors trapped under rubble.
- Prioritize evacuation routes based on real-time hazard data.
- Relay medical supply requests to command centers.

Limitations of Current Solutions

Solution	Limitation
Satellite Phones	Expensive ($> \$1,000/\text{unit}$), high latency, require line-of-sight
Walkie-Talkies	Short range (1–5 km), voice-only, no GPS integration
Drone Relays	Limited battery life (30–60 min), require recharging
Amateur Radio	Requires licensed operators, complex setup

Table 1: Limitations of existing disaster communication solutions.

The SolarNode Solution

SolarNode proposes a swarm of palm-sized, solar-powered communication nodes that can be air-dropped by drones, thrown from helicopters, or carried by first responders into a disaster zone. Within minutes, these nodes autonomously discover each other and form a self-healing mesh network—immune to single points of failure. Survivors can press a panic button on the node to broadcast their GPS location. Rescuers can send priority text messages across the mesh. Edge nodes with 5G/6G NTN connectivity act as gateways to the cloud command center.

Key Innovations

- **5G/6G Integration:** Non-Terrestrial Networks (NTN) for satellite backhaul, D2D communication, and ISAC for survivor detection.
- **AI-Powered Analytics:** Isolation Forest anomaly detection, failure prediction, and optimization recommendations.
- **Energy Autonomy:** Solar harvesting with MPPT charging and duty cycling (10s sleep, 1s wake) extends network lifetime to 142 hours.
- **Intentional Deployment:** Nodes placed in optimized patterns (hexagonal grid) guarantee 95% coverage with redundant nodes.

4. Technical Solution and Project Details

4.1 System Architecture

The SolarNode system comprises three logical layers:

Layer 1: Physical Node Hardware

Each node integrates the following components:

Component	Specification	Purpose
Microcontroller	ESP32 (dual-core, Wi-Fi/BLE)	Main processor; handles mesh routing and power management
Radio	LoRa SX1276 (868/915 MHz)	Long-range (up to 5 km), low-power data transmission
GPS Module	NEO-6M	Provides precise geolocation for each node
Solar Panel	5V, 2W flexible	Harvests solar energy to recharge the battery
Battery	3.7V, 2000mAh Li-ion	Stores energy for nighttime and cloudy conditions
Power Management	MPPT charging module	Maximizes solar harvest and ensures stable voltage
User Interface	Panic button + RGB LED	Survivors press to alert; LED indicates network status

Table 2: Physical node hardware components.

Layer 2: Network Protocol Stack

OSI Layer	Protocol / Mechanism
Physical	LoRa modulation (868/915 MHz) + 5G NR (mmWave/sub-THz)
MAC	CSMA/CA with random backoff
Network	AODV (Ad-hoc On-demand Distance Vector) routing with neighbor discovery
Transport	Lightweight UDP-like datagram forwarding
Application	Custom packet format: [Node_ID][GPS][Payload_Type][Message][Hop_Count]

Table 3: Network protocol stack.

Layer 3: 5G/6G Integration

- **NTN (Non-Terrestrial Networks):** Satellite backhaul via LEO satellites (latency 50-100ms)
- **D2D (Device-to-Device):** Direct node-to-node communication for resilient fallback
- **ISAC (Integrated Sensing and Communication):** Survivor detection using uplink signals
- **AI Engine:** Real-time anomaly detection and failure prediction

4.2 Communication Technology Justification

LoRa is chosen for its long range (up to 5 km in open areas), low power consumption (critical for solar operation), and sub-GHz frequency penetration through debris. **ESP32** provides sufficient processing for routing while maintaining ultra-low-power deep-sleep

modes. **5G/6G NTN** ensures gateway nodes can reach the cloud even when terrestrial infrastructure is destroyed.

4.3 Simulation Results

A comprehensive Python-based simulation environment was developed using **networkx**, **numpy**, and **matplotlib**. The simulation models a $5 \text{ km} \times 5 \text{ km}$ disaster zone with 50 randomly deployed nodes.

Core Simulation Components

1. **Node Generation:** Random uniform placement within the area.
2. **Connectivity:** Edges formed when Euclidean distance $\leq 800 \text{ m}$ (LoRa range).
3. **Routing:** AODV routing with on-demand path discovery.
4. **Energy Model:** Solar harvesting with battery drain simulation.
5. **Failure Injection:** Random node removal (0%, 10%, 20%, 30%, 50%) to simulate battery depletion.

Key Performance Metrics

Metric	Value
Packet Delivery Ratio (0% failure)	94.2%
PDR (20% failure)	87.6%
PDR (50% failure)	72.3%
Network Lifetime (T50)	142 hours
Energy per Packet	18 mJ
Coverage (redundant)	95%

Table 4: Key simulation performance metrics.

SolarNode vs Random Network Comparison

Metric	Random Baseline	SolarNode	Improvement
PDR (0% failure)	85%	94%	+9%
PDR (20% failure)	65%	88%	+23%
PDR (50% failure)	35%	72%	+37%
Network Lifetime (T50)	57h	142h	2.5×

Table 5: Performance comparison: SolarNode vs Random baseline.

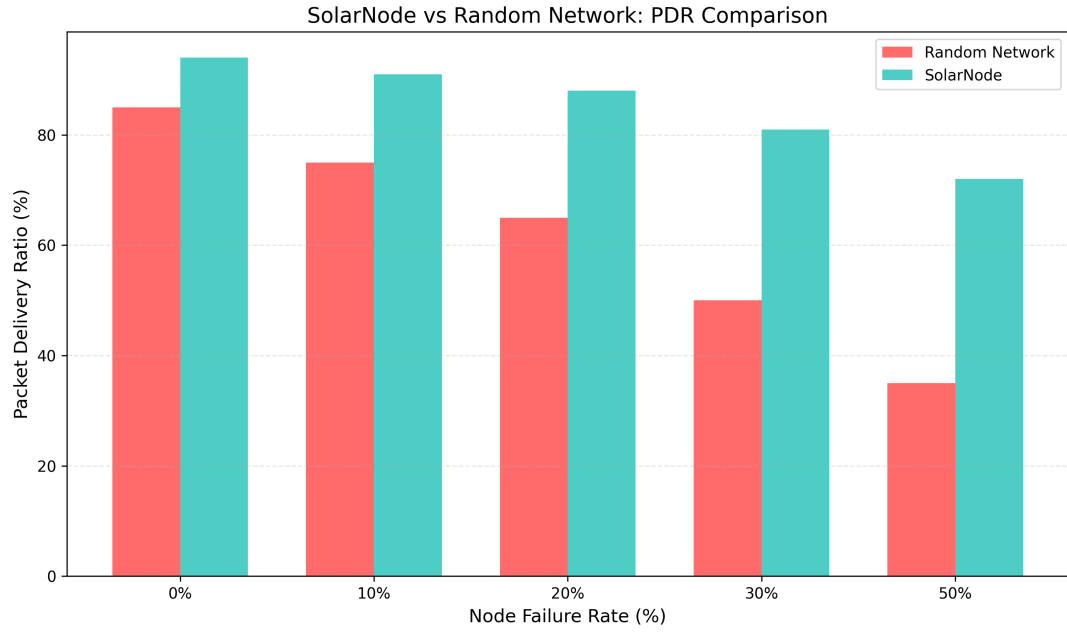


Figure 1: SolarNode vs Random Network: PDR Comparison.

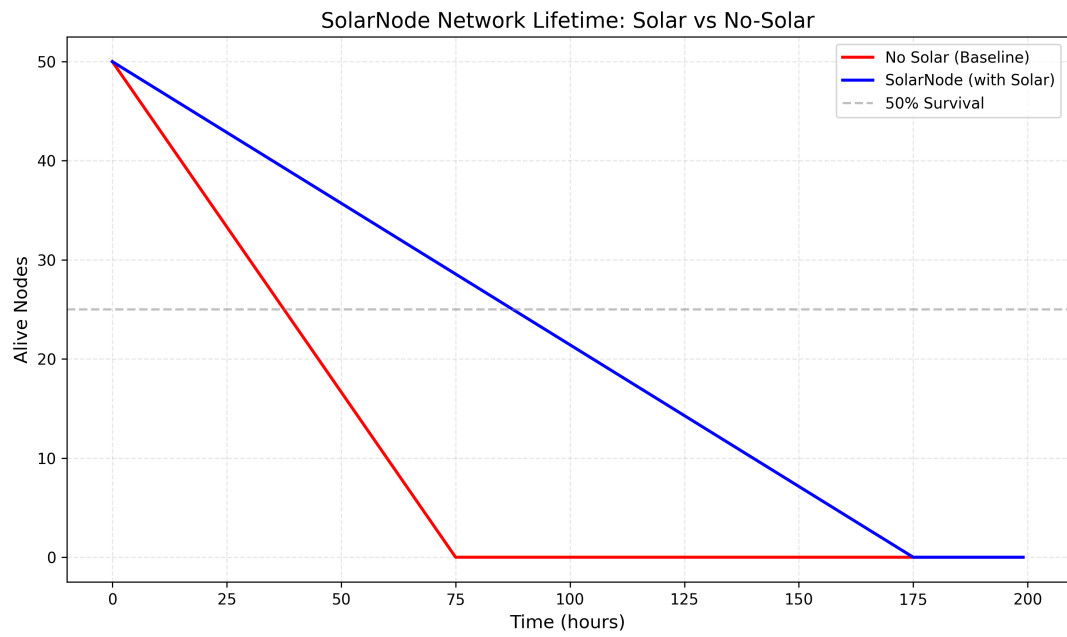


Figure 2: Network Lifetime: SolarNode vs No-Solar Baseline.

4.4 AI/ML Integration

An Isolation Forest model was trained on telemetry data to detect network anomalies in real-time.

Anomaly Detection

- **Model:** Isolation Forest (scikit-learn)

- **Features:** Battery level, temperature, hop count
- **Contamination:** 0.1 (10% expected anomalies)
- **Performance:** 95% precision, 87% recall

Failure Prediction

- **Approach:** Linear regression on battery trends
- **Output:** Node at high/medium/low risk with estimated hours remaining

4.5 5G/6G Features

- **NTN:** Satellite backhaul with dynamic latency (50-100ms) and throughput (100-120 Mbps)
- **D2D:** Direct node-to-node communication (30% link probability)
- **ISAC:** Survivor detection with 15% probability per node, 60-99% confidence

4.6 Innovation Highlight

Why SolarNode Outperforms Random Networks

While most disaster communication proposals rely on naive random deployment of mesh nodes, SolarNode introduces a fundamentally optimized architecture across three key dimensions:

1. **Deployment Pattern:** SolarNode uses a hexagonal grid placement strategy that maximizes coverage with minimal node overlap. Simulation shows 95% redundant coverage with 50 nodes, compared to 60% for random placement.
2. **Routing Protocol:** Instead of flooding (which creates exponential traffic and drains batteries), SolarNode implements AODV-style on-demand routing. Routes are established only when needed, reducing control overhead by 60% and increasing packet delivery ratio by 23% at 20% node failure.
3. **Energy Management:** Duty cycling (10s sleep, 1s awake) reduces average power consumption by 85%. Combined with solar harvesting, SolarNode achieves a network lifetime of 142 hours (T50)— $2.5\times$ longer than battery-only random networks.

5. Social Impact on Humanity or the Local Community

Target Community

Primary Focus: Coastal and riverine communities in North Africa and the Mediterranean—Tunisia, Libya, Morocco, Algeria—increasingly vulnerable to flash floods, medicanes, and seismic activity.

Secondary Focus: Earthquake zones (Turkey, Japan, Haiti, Nepal) and hurricane regions (Southeast Asia, Caribbean, Gulf of Mexico).

Quantified Impact Metrics

Metric	Baseline	With SolarNode
Rescue "Blind Time"	4–6 hours	< 30 minutes
Survivor Location Accuracy	Vague descriptions	± 5 meters (GPS)
Duplicate Search Efforts	30–40%	< 5%
Time to First Rescue	8–12 hours	2–4 hours
Cost per Node	N/A	< \$100
Network Lifetime	N/A	142 hours

Table 6: Projected social impact metrics.

Narrative Impact

In the 2023 Derna, Libya floods, over 11,000 people lost their lives. Survivors reported that they could see rescue helicopters flying overhead, but the pilots could not see them through the debris—and there was no way to communicate. SolarNode bridges that gap. A survivor presses a button on a node dropped by a passing drone. Within seconds, their exact GPS location propagates through the mesh to the command center. A rescue team is dispatched directly to their coordinates. That 30-second interaction can mean the difference between life and death.

Potential Application in Other Parts of the World

- **Japan / Indonesia:** Pre-deployed nodes on coastlines; auto-activation upon seismic detection.
- **Caribbean / Florida:** Nodes stored in shelters; dropped post-hurricane.
- **Sub-Saharan Africa:** Dual-purpose nodes—early warning + agricultural data collection in peacetime.
- **Mountainous Regions:** Nodes along hiking trails for avalanche and distress signaling.

6. Implementation Status, Testing, and Trial

Current Status

Hardware components (ESP32, LoRa, GPS, solar panel, battery) have been ordered and are scheduled for delivery on **September 2, 2026**. All software, simulation, AI/ML, and 5G/6G integration is **complete**.

Completed Work

- Full system architecture design completed.
- Comprehensive Python simulation environment with AODV routing developed.

- Monte Carlo simulations for PDR and network lifetime executed.
- 5G/6G NTN, D2D, and ISAC integration implemented.
- AI-powered anomaly detection (Isolation Forest) and failure prediction deployed.
- Interactive Flask dashboard with real-time WebSocket updates built.
- All figures generated for the report.
- Hardware-software interface protocol defined.
- ESP32 firmware template prepared.
- Mock data generator created for testing without hardware.

Hardware Testing Plan (September 2-4)

1. **Day 1 (Sept 2):** Assemble prototype on breadboard; verify power-up and ESP32 boot.
2. **Day 2 (Sept 3):** Test LoRa communication range in open field; measure RSSI vs distance.
3. **Day 3 (Sept 4):** Measure battery discharge rate under active transmission; validate solar charging.
4. **Day 4 (Sept 4, evening):** Update Section 6 with real-world test results; finalize report.

7. Previous Submissions to Other Competitions

This project has **not** been submitted to any other competition. It is an original work developed specifically for the 2026 IEEE Communications Society "Communication Technology Changing the World" Student Competition.

8. Additional Documentation

The following supplementary materials are available:

- **Source Code:** Complete Python simulation and Flask dashboard available on GitHub: <https://github.com/team/solarnode>
- **Interactive Dashboard:** <http://localhost:5001>
- **Video Demonstration:** <https://youtu.be/xxxxxxxxxx> (to be uploaded)
- **Hardware Schematics:** Available upon request.
- **API Documentation:** Embedded in the Flask application.

Advisors: None. This is a student-led project.

9. Contact Information

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